

William Sander

Software Engineering | Applied Machine Learning

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Professional Summary

Production-oriented Software Engineer operating at the intersection of systems and applied ML. Built and deployed automation systems and software supporting large-scale operations and developed multilingual deep learning models from raw data to live inference. Comfortable owning ambiguous problems end-to-end across software, data workflows, and hardware systems.

Education

University of Vermont

B.S. Computer Science

Burlington, VT

Expected May 2026

Relevant Coursework: Advanced Machine Learning, Advanced Programming (C++), Algorithms, Data Structures, Database Systems, Operating Systems

Projects

AI Speech Emotion Recognition System

UVM, Advanced Machine Learning

PyTorch / Python / Deep Learning / ML Evaluation / AI Training

[Speech Emotion Recognition \(CRNN\) — GitHub](#)

Designed, trained, and evaluated a multilingual end-to-end deep learning speech emotion recognition neural network.

- Built a convolutional recurrent neural network (CRNN) combining CNNs and BiLSTMs with attention, achieving **81.29% accuracy and 83.63% F1** (4-class) and **76.76% accuracy and 82.80% F1** (6-class) across **8 multilingual datasets** (~25.8k samples)
- Added **distance-weighted loss** using arousal-valence theory and multilingual training; compared performance against classical ML baselines (253-feature extraction, ensemble models)
- Built modular data pipeline & interpretability tools (attention maps, Grad-CAM); deployed Flask live inference (**10-20ms latency**)

C++ Canvas Raster Graphics Editor

UVM, Advanced Programming

C++ / OpenGL / Command Pattern / NumPy

[Video Demo - YouTube](#) · [GitHub Repository](#)

- Built real-time raster graphics engine app with **7 tools**, multi-layer RGBA buffers, and **diff-based undo/redo**; custom rasterization, per-pixel alpha compositing, and zero-copy buffer reuse keep total memory under **140 MB**
- Implemented a real-time OpenGL pipeline with **lazy CPU-to-GPU texture updates** and software blending, **60 FPS** at **800×600**; Gaussian/Sobel/unsharp filters via **Python C API + NumPy** with direct C++↔Python data conversion.

E-Ink iCloud Calendar Display

IoT / OOP / Python / 3D Printing / FreeCAD / Hardware

[GitHub Repository](#)

- Developed an IoT E-Ink display calendar product, integrating iCloud and public holiday API, merging ICS/JSON events
- Engineered a modular, object-oriented software architecture with custom layout engine and competitive materials cost.

Work Experience

IT Clerk — Automation & Software Development

Burlington, VT

U.S. Census Bureau

Feb 2020–Nov 2020

Hired as IT Clerk; transitioned into an automation-focused software development role. Designed, built, and deployed production automation systems in a high-security federal environment. Instrumented and benchmarked workflows against historical completion times, eliminating manual processing bottlenecks and improving system reliability during large-scale field operations.

- Designed and implemented internal automation software & systems in **Python, C++, and SQL**, eliminating **30+ hours/week** of manual reconciliation work (**1,000+ hours annually**)
- Architected and deployed production software under real-time operational constraints during COVID-19 census extension
- Developed validation and consistency systems to detect & correct device custody and configuration issues across distributed systems
- Diagnosed and resolved large-scale mobile device and iOS issues impacting **hundreds of enumerators**, improving operational uptime
- Supported authentication, access control, and compliance across **3,000+ mobile devices** in coordination with regional IT teams

Skills

Languages: Python, C++, C, Java, R, SQL

Major Systems & Tools: Git, Cursor, VSCode, Docker, Linux, REST APIs, PyTorch, Flask

Volunteer Experience

Volunteer Tech Instructor

Technology for Tomorrow

- Taught computer skills and provided technical support to community members, improving digital literacy and access